

JEE ADVANCED WEEKEND TEST SERIES

44
YEARS
OF EXCELLENCE



NARAYANA
IIT-JEE ACADEMY-INDIA

ISR_*CO-SC(MODEL-B)

Time: 3 Hrs

Date: 07-04-24

Max. Marks: 186

07-04-24_ISR.IIT_STAR CO-SC(MODEL-B)_JEE ADV_WAT-40_SYLLABUS

MATHEMATICS: **PRESENT WEEK(100%):** Revision of Circles/Parabola/Ellipse/ Hyperbola

PHYSICS: **PRESENT WEEK(80%):** Sound waves (longitudinal waves) as pressure waves, displacement waves, velocity of sound waves and echoes, Power, Intensity in sound waves, Organ pipes, Experiment: Speed of sound using resonance
PREVIOUS WEEK(20%): Waves & Sound: Introduction to wave motion, classifications, equation of Progressive plane waves and pulses, Transverse waves on stretched strings velocity, power transmitted, Energy Superposition (Interference) of waves, Reflection and transmission of waves at a boundary, Standing waves & modes of vibration of stretched strings, energy and power in standing waves. (Exclude: Sound Wave)

CHEMISTRY: **PRESENT WEEK(80%): Colloids:** types, methods of preparation and general properties, Elementary ideas of emulsions, surfactants and micelles (only definitions and examples). Applications of colloids (Exclude: Catalysis)

PREVIOUS WEEK(20%): SOLID STATE: Crystalline, amorphous solids, types of crystalline solids. Primitive cubic, body central cubic, face centered cubic, close packing of atoms, CCP, HCP, Hexagonal primitive radius ratio, Type of ionic solids, Crystal defects (Exclude spinel, inverse spinel, miller indices, Electrical & magnetic properties of solids)

SURFACE CHEMISTRY: Physisorption and Chemisorption, Freundlich adsorption isotherm (Exclude: Langmuir adsorption isotherm)

MATHS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 5 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	8	32
Sec – III (Q.N : 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	6	18
Total				18	62

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 22)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 23 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	8	32
Sec – III (Q.N : 31 – 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	6	18
Total				18	62

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 40)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 41 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	8	32
Sec – III (Q.N : 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	6	18
Total				18	62

SECTION - I
(SINGLE CORRECT ANSWER TYPE)

This section contains **4** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

- Three circles touch each other externally. The tangents at their points of contact meet at point whose distance from the point of contact is 4. The ratio of the product of the radii to the sum of the radii of the circle is
A) 16 : 1 B) 16 : 3 C) 16 : 5 D) 16 : 7
- The mirror image of the parabola $y^2 = 4x$ in the tangent to the parabola at the point (1, 2) is
A) $(x-1)^2 = 4(y+1)$ B) $(x+1)^2 = 4(y+1)$ C) $(x+1)^2 = 4(y-1)$ D) $(x-1)^2 = 4(y-1)$
- If $y = x$ and $3y + 2x = 0$ are the equations of a pair of conjugate diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then its eccentricity is
A) 1/2 B) 1/3 C) $1/\sqrt{3}$ D) $\sqrt{3}/2$
- The equation of the common tangent to the curve $y^2 = 8x$ and $xy = -1$ is
A) $3y = 9x + 2$ B) $y = 2x + 1$ C) $2y = x + 8$ D) $y = x + 2$

SECTION - II
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE or More is/are** correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- The circles $x^2 + y^2 + 2x + 4y - 20 = 0$ and $x^2 + y^2 + 6x - 8y + 10 = 0$
A) are such that the number of common tangents on them is 2
B) are orthogonal
C) are such that the length of their common tangent is $5(12/5)^{1/4}$
D) are such that the length of their common chord is $5\sqrt{3/2}$
- Let P be a point on the circle $x^2 + y^2 = 9$, Q a point on the line $7x + y + 3 = 0$, and the perpendicular bisector of PQ be the line $x - y + 1 = 0$. Then the coordinates of P are
A) (3,0) B) (0,3) C) $\left(\frac{72}{25}, -\frac{21}{25}\right)$ D) $\left(-\frac{72}{25}, \frac{21}{25}\right)$

7. Let the equations of a circle and parabola be $x^2 + y^2 - 4x - 6 = 0$ and $y^2 = 9x$ respectively. Then
- A) (1, -1) is a point on the common chord of contact
B) the equation of the common chord is $y+1=0$
C) the length of the common chord is 6
D) None of these
8. Tangents are drawn from point $(-2,0)$ to $y^2 = 8x$, radius of circle(s) that would touch these tangents and the corresponding chord of contact, can be equal to
- A) $4(\sqrt{2}+1)$ B) $4(\sqrt{2}-1)$ C) $8\sqrt{2}$ D) $4\sqrt{2}$
9. Let F_1, F_2 be two foci of the ellipse and PT and PN be the tangent and the normal respectively to the ellipse at point P then
- A) PN bisects $\angle F_1PF_2$ B) PT bisects $\angle F_1PF_2$
C) PT bisects angle $(180^\circ - \angle F_1PF_2)$ D) None of these
10. Let A and B be two fixed (distinct) points and P be another point in the same plane, which moves in such a way that $\lambda_1 PA + \lambda_2 PB = \lambda_3$ where λ_1, λ_2 and λ_3 are real constants. The locus of P is
- A) a circle if $\lambda_1 = 0$ and $\lambda_2 \lambda_3 > 0$ B) a circle if $\lambda_1 > 0, \lambda_2 < 0$ and $\lambda_3 = 0$
C) an ellipse if $\lambda_1 = \lambda_2 > 0$ and $\lambda_3 \lambda_1 > AB$ D) a hyperbola if $\lambda_2 = -1$ and $\lambda_1 \lambda_3 > 0$
11. If the ellipse $x^2 + k^2 y^2 = k^2 a^2; k^2 > 1$ is confocal with the hyperbola $x^2 - y^2 = a^2$, then
- A) ratio of eccentricities of ellipse & hyperbola is $1/\sqrt{3}$
B) ratio of major axis of ellipse and transverse axis of hyperbola is $\sqrt{3}$
C) The ellipse and hyperbola cuts each other orthogonally
D) ratio of length of latus rectum of ellipse and hyperbola is $1/3$
12. For the hyperbola $9x^2 - 16y^2 - 18x + 32y - 151 = 0$
- A) one of the directrix is $x = \frac{21}{5}$ B) Length of latus rectum = $9/2$
C) foci are (6, 1) and (-4, 1) D) eccentricity is $5/4$.

SECTION-3
(NUMERICAL VALUE ANSWER TYPE)

This section contains **6** questions. The answer to each question is a Numerical Values comprising of positive or negative decimal numbers (**Place value ranging from Thousands place to Hundreds place**).

Eg. 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc

Marking scheme: +3 for correct answer, in all other cases.

13. Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres $(-3, 0), (-1, 0), (1, 0)$ and $(3, 0)$ respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C . If the length of this chord can be expressed as $\sqrt{l}$, then the value of $\sqrt{l}$ is
14. If the curves $ax^2 + 4xy + 2y^2 + x + y + 5 = 0$ and $ax^2 + 6xy + 5y^2 + 2x + 3y + 8 = 0$ intersect at four concyclic points, then the value of $-\frac{a}{8}$ is
15. The straight line $x + y = k$ touches the parabola $y = x - x^2$, then the value of $\frac{k}{8}$ is
16. If A and B are points on the parabola $y^2 = 4ax$ with vertex O such that OA is perpendicular to OB and having lengths r_1 and r_2 respectively, then the value of $\frac{1}{5} \frac{r_1^{4/3} r_2^{4/3}}{(r_1^{2/3} + r_2^{2/3}) a^2}$ is
17. If the normal to the rectangular hyperbola $xy = c^2$ at any point ' t ' on it intersects the hyperbola at ' t_1 ' then the value of $\frac{t^6 t_1^2}{4}$ is
18. A straight line PQ touches the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the circle $x^2 + y^2 = r^2$ ($3 < r < 4$). RS is a focal chord of the ellipse which is parallel to PQ and meets the circle at points R and S , then the value of $\frac{RS}{4}$ is

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

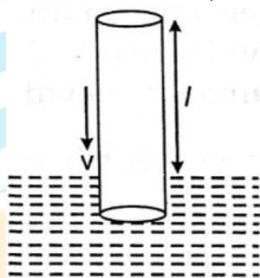
This section contains **4 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct**.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

19. The specific gravity of oxygen and nitrogen are in the ratio of 16 : 14. The temperature at which the velocity of sound will be the same as that of nitrogen at 15°C, will be:

A) 112°C B) 72°C C) 48° D) 56°C

20. An open pipe of sufficient length is dipping in water with a speed V vertically. If at any instant l is length of tube above water then the rate at which fundamental frequency of pipe changes is: (C is the speed of sound in air)



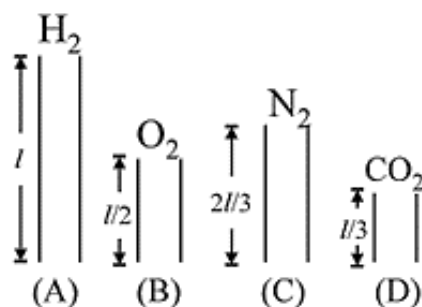
A) $\frac{CV}{2l^2}$ B) $\frac{CV}{4l^2}$ C) $\frac{CV}{2v^2l^2}$ D) $\frac{CV}{4v^2l^2}$

21. A simple harmonic oscillator of frequency ' f ' is attached to the end of a cord that has a linear mass density ' μ ' and is under a tension ' T '. The average power that must be provided to the cord by the oscillator to generate a sinusoidal wave of amplitude ' A ', angular frequency ' ω ' and speed ' v ' is

A) $\mu\omega^2 A^2 v$ B) $\frac{\mu\omega^2 A^2 v}{2}$ C) $\frac{\mu\omega^2 A^2 v^2}{2}$ D) $\frac{3\mu\omega^2 A^2 v}{2}$

22. Four open organ pipes of different lengths and different gases at same temperature are shown in figure. Let f_A , f_B , f_C and f_D be their fundamental frequencies then:

[Take $\gamma_{CO_2} = 7/5$]



A) $f_D/f_A = \sqrt{76/11}$

B) $f_B/f_C = \sqrt{72/28}$

C) $f_C/f_D = \sqrt{11/28}$

D) $f_A/f_B = \sqrt{2}$

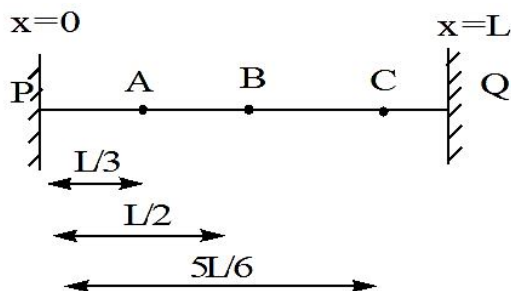
SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

23. Let a disturbance y be propagated as a plane wave along the x -axis. The wave profiles at the instants $t = t_1$ and $t = t_2$ are represented respectively as $y_1 = f(x_1 - vt_1)$ and $y_2 = f(x_2 - vt_2)$. The wave propagating without change of shape.
- A) The velocity of the wave is v .
- B) The velocity of the wave is $v = (x_2 + x_1)/(t_2 + t_1)$
- C) The particle velocity is $v_p = -vf'(x - vt)$
- D) The **phase** velocity is v .
24. Two particles of a medium disturbed by the wave propagation are at $x_1 = 0$ and $x_2 = 1$ cm. The wave is propagating in positive x -direction. The displacement of the particles is given by the equation: $y_1 = 2(\sin 3\pi t)$ cm and $y_2 = 2\sin(3\pi t - \pi/8)$ cm (t is in seconds)
- A) The frequency of wave is 1.5 Hz.
- B) Wavelength of the wave can be 16 cm.
- C) Velocity of the wave can be 24 cm/s.
- D) Wave equation can be $y = (2)\sin\left[\frac{2\pi}{16}(24t - x)\right]$ cm.
25. Standing waves are established on a string of length L such that A is a node and B is an immediate antinode. Oscillation amplitude for point B is a_0 . Let a' be the oscillation amplitude for point C. Pick the suitable option(s) for correct value of a' and the possible equation(s) for standing wave on the string.



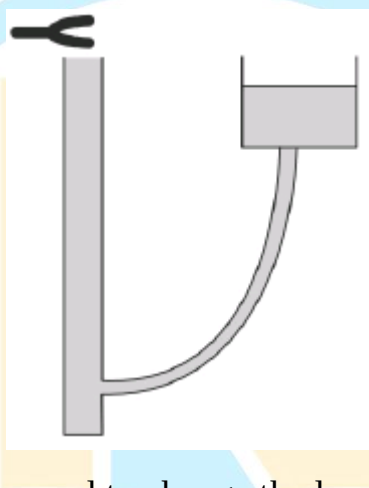
A) $a' = \frac{a_0}{2}$

B) $y = a_0 \sin\left(\frac{6\pi}{L}x\right)\cos(\omega t)$

C) $a' = a_0$

D) $y = a_0 \sin\left(\frac{3\pi x}{L}\right)\sin(\omega t)$

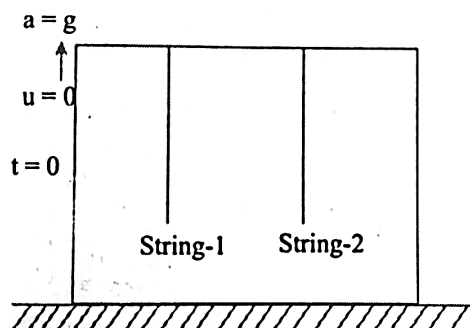
26. In resonance column experiment a tuning fork of frequency $f = 400\text{Hz}$ is held above the pipe as shown in figure.



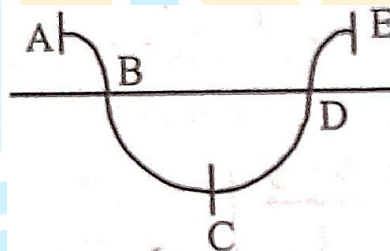
The reservoir is raised and lowered to change the level of water and thus the length of the column of air in the tube. The area of cross section of the reservoir is 6 times that of the pipe. Initially, the reservoir is kept so that the pipe is full up to the brim. Tuning fork is sounded and the reservoir is lowered. When the reservoir is lowered by 21 cm, first resonance is recorded. When the reservoir is lowered further by 49 cm the second resonance is heard.

- A) Wave length is $\lambda = 84\text{cm}$
 B) Wave length is $\lambda = 75\text{cm}$
 C) Velocity of sound is 300 m/s
 D) Velocity of sound is 336 m/s

27. Two identical strings each of mass m and length ℓ are connected on ceiling of a lift as shown in figure. Two wave pulses one at upper end of string-1 and other at lower end of string-2 are produced at simultaneously at $t = 0$ and lift also starts moving from ground upwards with constant acceleration g at same instant. Choose the correct option(s)



- A) Both the wave pulses will be at same horizontal level at $t = \sqrt{\frac{\ell}{2g}}$
- B) Acceleration of pulse in string-1 with respect to ground is $2g$ downwards
- C) Acceleration of pulse in string-2 with respect to ground is g upwards
- D) Height raised by lift when both pulses are at same horizontal level is $\frac{\ell}{4}$
28. The figure represents a longitudinal wave travelling in positive x -direction. Then



- A) part ABC represents compression B) part ABC represents rarefaction
- C) part CDE represents compression D) part CDE represents rarefaction
29. An air column in a pipe closed at one end is made to vibrate in its second overtone by a tuning fork of frequency 440Hz . The speed of sound wave in air is 330 m/s . end corrections may be neglected. Let P_0 denote the mean pressure at any point in the pipe and ΔP_0 the amplitude of pressure variation. Then:
- A) Length of the pipe is $\frac{15}{16}m$ B) Length of the pipe is $\frac{9}{16}m$
- C) The pressure at the open end is P_0 D) The **excess** pressure at the open end is P_0

30. In a resonance tube experiment, a closed organ pipe of length 120 cm resonates when tuned with a tuning fork of frequency 340Hz. If water is poured in the pipe then (velocity of sound in air is 340 m/s)
- A) Minimum length of water column to have the resonance is 45cm
- B) The distance between two successive nodes is 50cm
- C) The maximum length of water column to create the resonance is 95cm
- D) There will be 3 resonances are possible

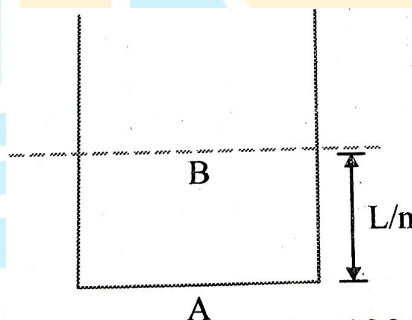
SECTION – III (NUMERICAL VALUE ANSWER TYPE)

This section contains **6 questions**. The answer to each question is a **Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place)**.

Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

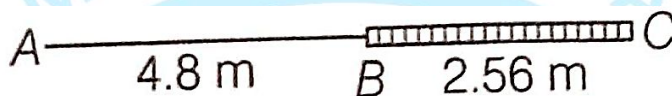
Marking scheme : +3 for correct answer, 0 in all other cases.

31. The average power transmitted across a cross-section by two sound waves moving in the same direction are equal. The wave lengths of two sound waves are in the ratio of 1 : 2, then find the ratio of their pressure amplitudes.
32. A closed organ pipe of length L is vibrating in fundamental frequency. There are two points A and B in the organ pipe as shown, at a distance $AB = L/n$. Ratio of Maximum pressure variation at point A to point B is $2/\sqrt{3}$ find value of n .



33. In the experiment for the determination of the speed of sound in air using the resonance column method, the length of the air column that resonates in the fundamental mode with a tuning fork is 0.1m. When this length is changed to 0.4m, the same tuning fork resonates with the first overtone. The end correction is (in m)
34. A string of length 0.3m and mass 10^{-2} kg is clamped at both its ends. The tension in the string is 1.2N. When a pulse travels along the string, the shape of the string is found to be the same at times t and $t + \Delta t$. The minimum value of Δt is $\frac{1}{2k}$ s. Find the value of k .

35. A transverse wave of amplitude $0.5 \times 10^{-3} \text{ m}$ and frequency $\frac{100}{\pi} \text{ Hz}$ is produced on a wire stretched to a tension of 100 N . The wave speed is 100 ms^{-1} . The average power that the source is transmitting to the wire is $\alpha \times 10^{-3} \text{ watt}$. Find the value of α .
36. A long wire ABC is made by joining two wires AB and BC of equal radii. AB has length 4.8 m and mass 0.06 kg . BC has length 2.56 m and mass 0.2 kg . The wire ABC is under a tension 80 N . A sinusoidal wave pulse of amplitude 3.5 cm is sent along the wire AB from the end A. The time taken by the wave pulse to reach the other end C of the wire is $\frac{2}{100} \alpha$ second. Find the value of α .

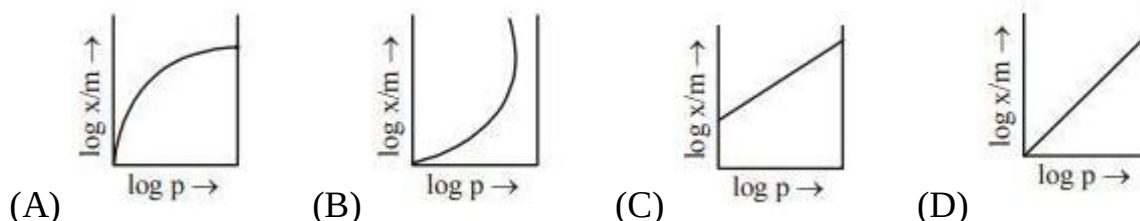
**CHEMISTRY****Max.Marks:62**

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct**.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

37. ΔG , ΔH and ΔS for adsorption of a gas on a solid absorbent are respectively
 (A) $-, +, -$ (B) $-, -, +$ (C) $-, -, -$ (D) $-, +, +$
38. The flocculation values of KCl , MgCl_2 , CrCl_3 and SnCl_4 for a negatively charged sol are in the order
 (A) $\text{KCl} < \text{MgCl}_2 < \text{CrCl}_3 < \text{SnCl}_4$ (B) $\text{KCl} = \text{MgCl}_2 = \text{CrCl}_3 = \text{SnCl}_4$
 (C) $\text{MgCl}_2 < \text{KCl} < \text{CrCl}_3 < \text{SnCl}_4$ (D) $\text{SnCl}_4 < \text{CrCl}_3 < \text{MgCl}_2 < \text{KCl}$
39. Which of the following curves is in according with Freundlich adsorption isotherm?



40. 1.30 cm^3 of N_2 gas at STP is absorbed per gram of silica gel. The area occupied by nitrogen molecule is 0.16 nm^2 . What is the surface area per gram of silica gel?
($N_A = 6.023 \times 10^{23}$)
- (A) $1.6 \text{ m}^2 \text{ g}^{-1}$ (B) $5.568 \text{ m}^2 \text{ g}^{-1}$ (C) $3.48 \text{ m}^2 \text{ g}^{-1}$ (D) $4.42 \text{ m}^2 \text{ g}^{-1}$

SECTION – II
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

41. Which of the following is/are lyophobic colloids?
(A) Gold sol (B) As_2S_3 sol (C) $\text{Fe}(\text{OH})_3$ sol (D) Starch sol
42. Select the correct statement(s)
(A) Brownian motion and Tyndall effect are shown by true solutions
(B) Sorption process is combination of adsorption and absorption process
(C) Hardy-Schulze law is related with coagulation of a sol
(D) Higher is the gold number lesser will be the protective power of a lyophilic colloid
43. Micelles are formed only
(A) below the CMC and the Kraft temperature
(B) above the CMC and below the Kraft temperature
(C) above the CMC and above the Kraft temperature
(D) below the CMC and above the Kraft temperature
44. Ions of NaCl which are touched by 1 body diagonal are
(A) Cl^- ions present at the corner of cube
(B) Cl^- ions present at the face center of cube
(C) Na^+ ions present at the edge center of cube
(D) Na^+ present at body center of cube

45. Which of the following statements is/are correct?
- (A) Dislocation of ion from lattice site to interstitial site is called Frenkel defect
- (B) Missing of positive and negative ions from their respective position producing a pair of hole's called Schottky defect
- (C) The presences of ions in the vacant interstitial sites along with lattice point is called interstitial defect
- (D) Non-stoichiometric NaCl is yellow solid
46. If 'a' is the edge length of the unit cell then for zinc blende structure the distance of closest approach between cation and anion is
- (A) $\frac{\sqrt{2}a}{2}$ (B) $\frac{\sqrt{3}a}{2}$ (C) $\frac{\sqrt{2}a}{4}$ (D) $\frac{\sqrt{3}a}{4}$
47. In a close packed structure of mixed oxides, the lattice is composed of oxide ions, one eighth of tetrahedral voids are occupied by divalent cations while one half of octahedral voids are occupied by trivalent cations. What is the formula of the oxide?
- (A) ABO_4 (B) AB_2O_4 (C) AB_2O (D) ABO
48. Which of the following is the application of adsorption?
- (A) Deionization of water (B) Gas masks
- (C) Hygroscopic nature of $CaCl_2$ (D) Heterogeneous catalysis

SECTION – III
(NUMERICAL VALUE ANSWER TYPE)

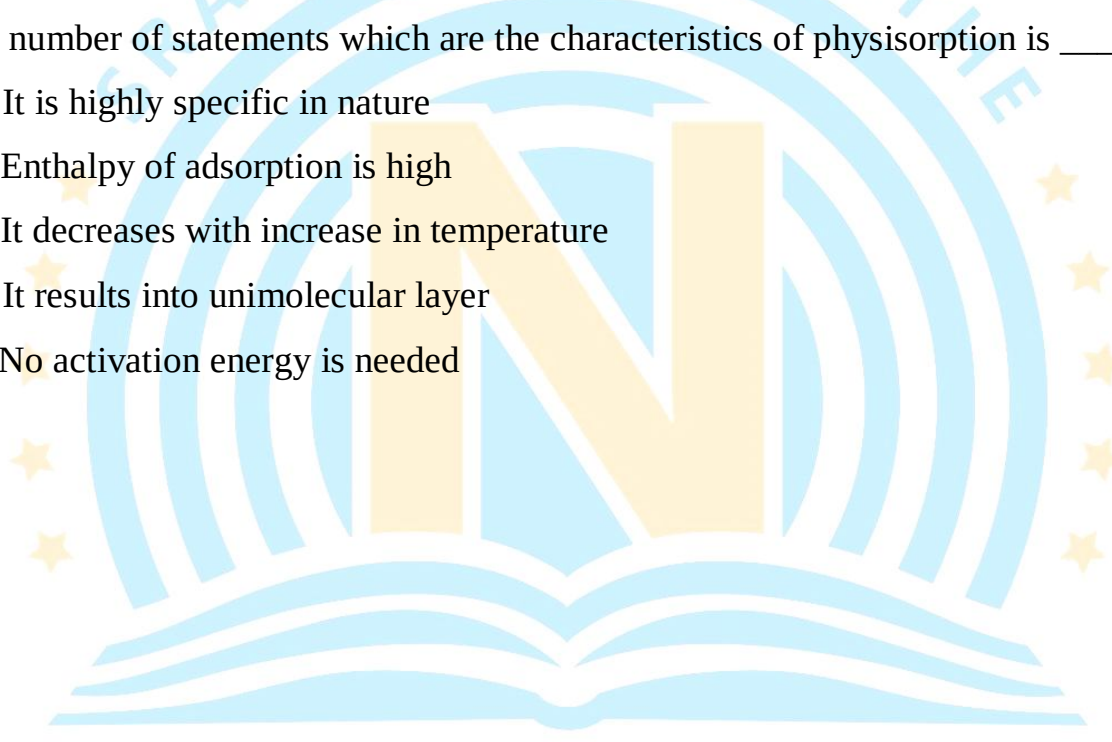
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Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme : +3 for correct answer, 0 in all other cases.

49. A detergent ($C_{12}H_{25}SO_4^-Na^+$) solution becomes a colloidal sol at a conc of $10^{-3}M$. On an average 10^{13} colloidal particles present in $1mm^3$. Then the average number of ions are contain in one colloidal particle micelle is
50. When 200 mL of 0.2 M acetic acid is shaken with 0.6g of wood charcoal, the final concentration of acetic acid after adsorption is 0.1 M. The mass of acetic acid adsorbed per gram of carbon is ____ g

51. The mass of gas adsorbed, 'x' per unit mass of adsorbate, 'm' was measured at various pressure, p. A graph between $\log \frac{x}{m}$ and $\log p$ gives a straight line with slope equal to 2 and the intercept equal to 0.4771. The value of $\frac{x}{m}$ at a pressure of 4 atm is
[Given: $\log 3 = 0.4771$]
52. 20 mL of 0.5 M NaCl is required to coagulate 200 mL of As_2S_3 solution in 2 hours. The coagulating values of NaCl is (in m mol) _____
53. The number of colloidal systems from the following which will have 'liquid' as the dispersion medium, is _____ gem stones, paints, smoke, cheese, milk, hair cream, insecticide sprays, froth, soap lather
54. The number of statements which are the characteristics of physisorption is _____
- (A) It is highly specific in nature
 - (B) Enthalpy of adsorption is high
 - (C) It decreases with increase in temperature
 - (D) It results into unimolecular layer
 - (E) No activation energy is needed



THE NARAYANA GROUP